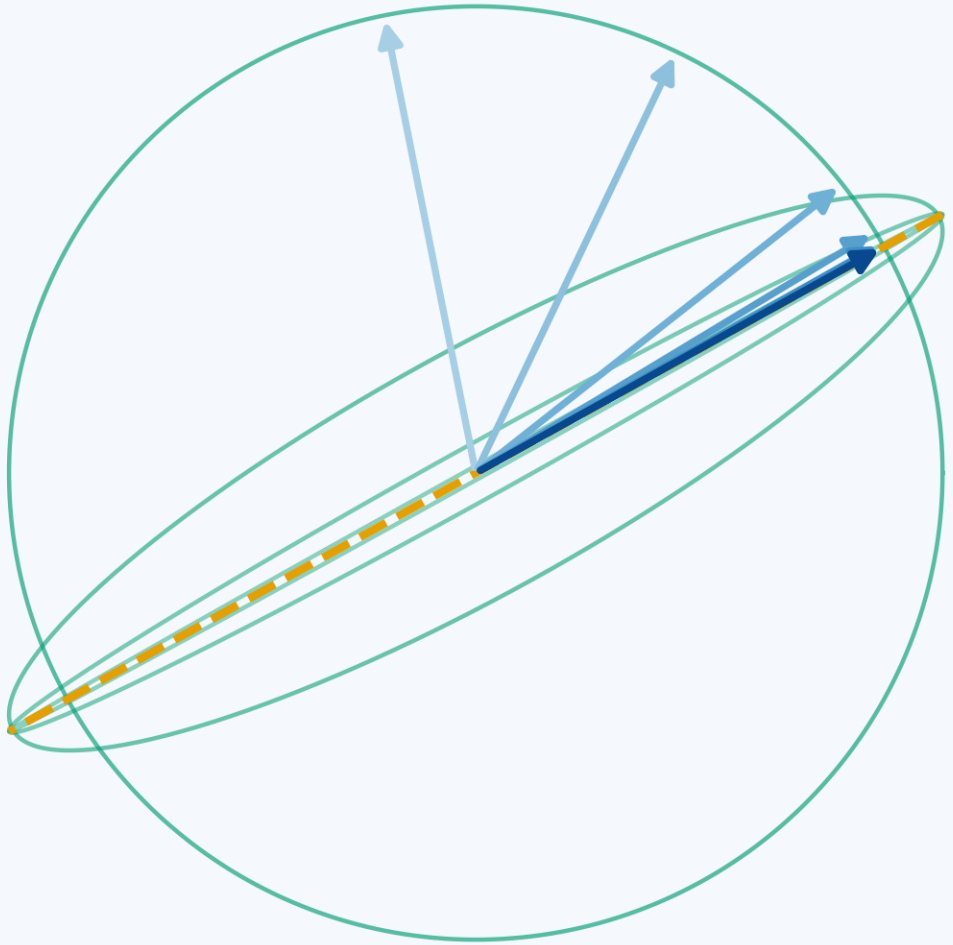
Eigenvalue Demos: Power, Inverse and QR Iteration

Companion to the interactive module



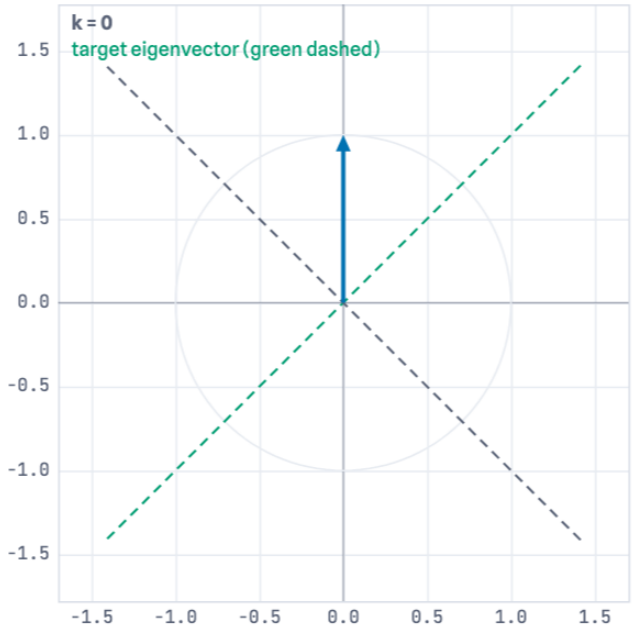
How to use the interactive module

Eigenvalue Problems

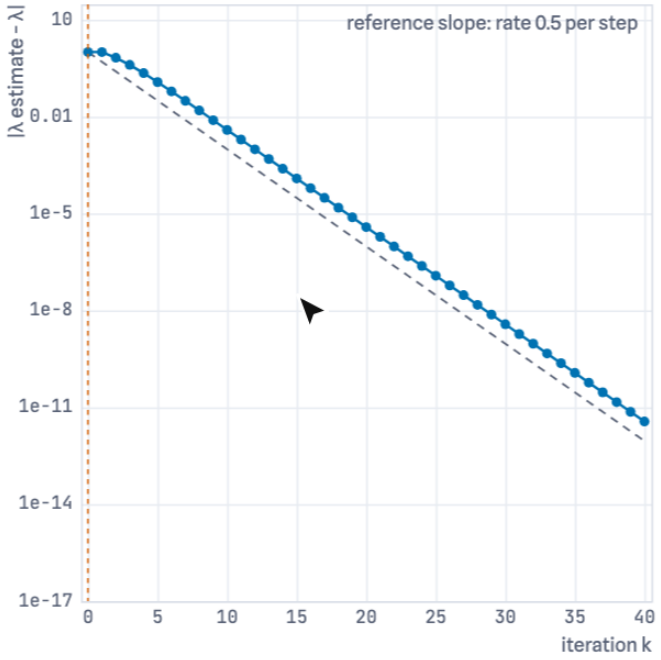
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Power and inverse iteration | QR iteration with shifts | Hessenberg reduction | Jacobi's method

ITERATES X_k (2 × 2 ONLY; DRAG THE START VECTOR)



ERROR IN THE EIGENVALUE ESTIMATE



Power and inverse iteration

Examples: choose an example...

Size n: 2 × 2

A:

3	1
1	3

Method: power iteration

Shift σ: 0

◀ Back | Step ▶ | +10 | Restart

Iterates

```
eigenvalues of A: 4, 2
 k   xₖ (max entry = 1)   λ estimate
 0          [0, 1]              3
```

Demo 1: Power and inverse iteration

Heath §4.5.1–4.5.3 · Algorithms 4.2–4.4 · Examples 4.10–4.14

Power iteration: multiply by A and rescale

$$y_k = A\,x_{k-1}, \qquad x_k = y_k / \|y_k\|_\infty$$

$$|\lambda_1| \approx \|y_k\|_\infty$$

$$\text{rate} = |\lambda_2 / \lambda_1|$$

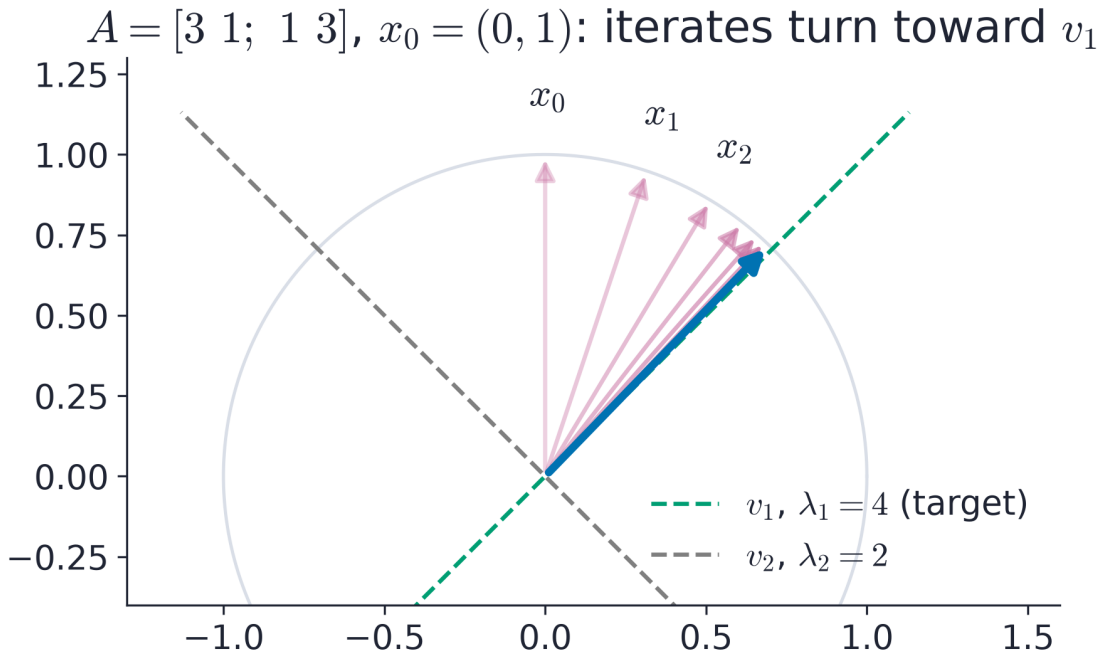
- $x_0 = \Sigma\, \alpha_j v_j$: A^k multiplies each part by λ_j^k, so the λ_1 part wins
- needs $|\lambda_1| > |\lambda_2|$ and $\alpha_1 \neq 0$; one matrix–vector product per step
- a better estimate: the Rayleigh quotient $x^T Ax \,/\, x^T x$

In the demo

- Method: power iteration
- 2 × 2: the arrows turn toward v_1 (green dashed)
- drag in the top plot to choose x_0
- error plot: dashed slope $|\lambda_2/\lambda_1|$ per step
- +10 jumps ahead; Back steps back

Power iteration finds the eigenvalue of largest magnitude and its eigenvector, and nothing else.

What the demo draws



Iterates as unit vectors. Green dashed: the target eigenvector; grey dashed: the other one.



Error versus k. The grey dashed reference has slope $|\lambda_2/\lambda_1| = 1/2$ per step.

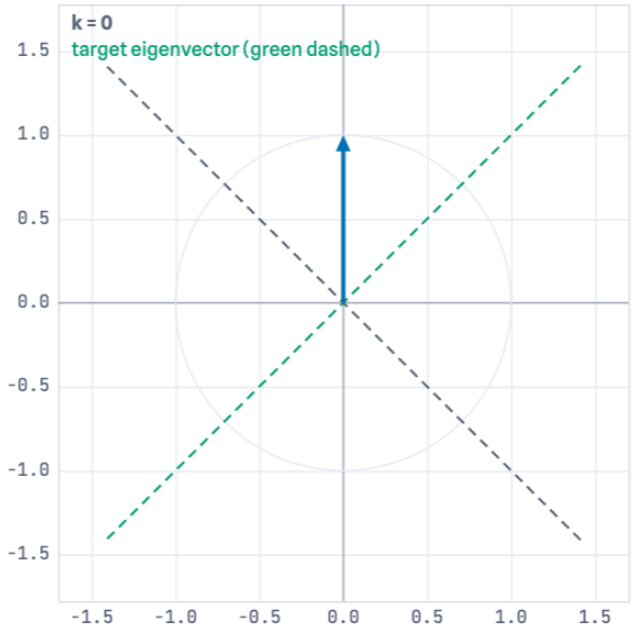
What the demo shows: Initial approximation

Eigenvalue Problems

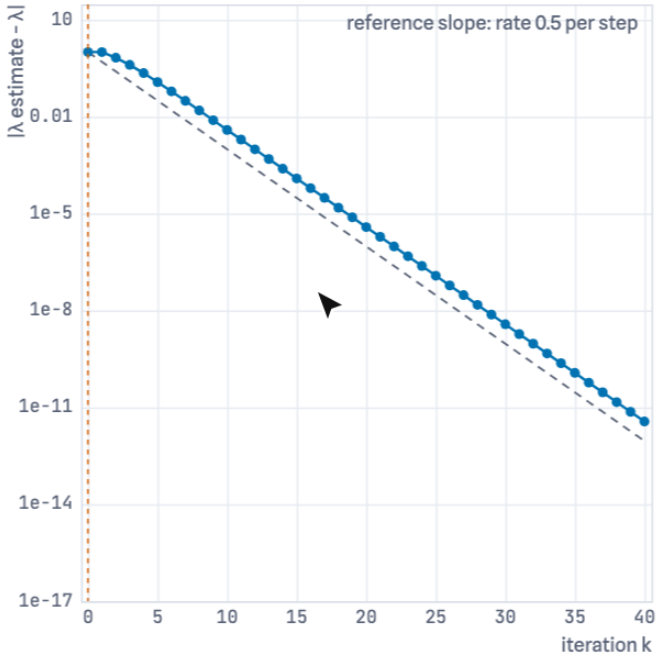
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ERROR IN THE EIGENVALUE ESTIMATE



Power and inverse iteration

Examples: choose an example...

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1	3

Method: power iteration

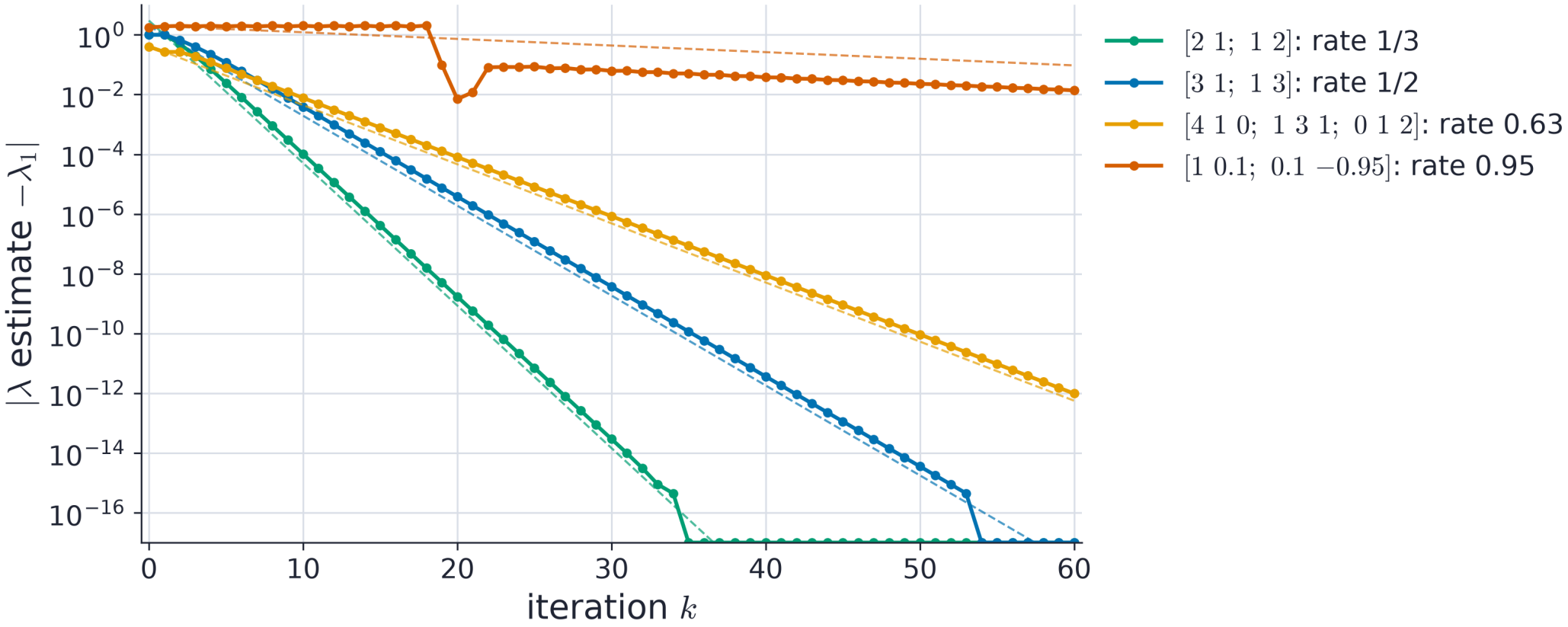
Shift σ: 0

◀ Back | Step ▶ | +10 | Restart

Iterates

```
eigenvalues of A: 4, 2
 k   xₖ (max entry = 1)   λ estimate
 0          [0, 1]              3
```

The rate depends only on $|\lambda_2 / \lambda_1|$



Linear convergence with constant $|\lambda_2/\lambda_1|$. When the top two eigenvalues are close in magnitude (0.955 vs 1.005) power iteration crawls: about 45 steps per decimal digit.

Inverse iteration: aim at the eigenvalue nearest σ

$$(A - \sigma I)\, y_k = x_{k-1}, \qquad x_k = y_k / \|y_k\|_\infty$$

$$|\lambda - \sigma| \approx 1 / \|y_k\|_\infty$$

$$\text{rate} = \frac{|\lambda_{\text{closest}} - \sigma|}{|\lambda_{\text{next}} - \sigma|}$$

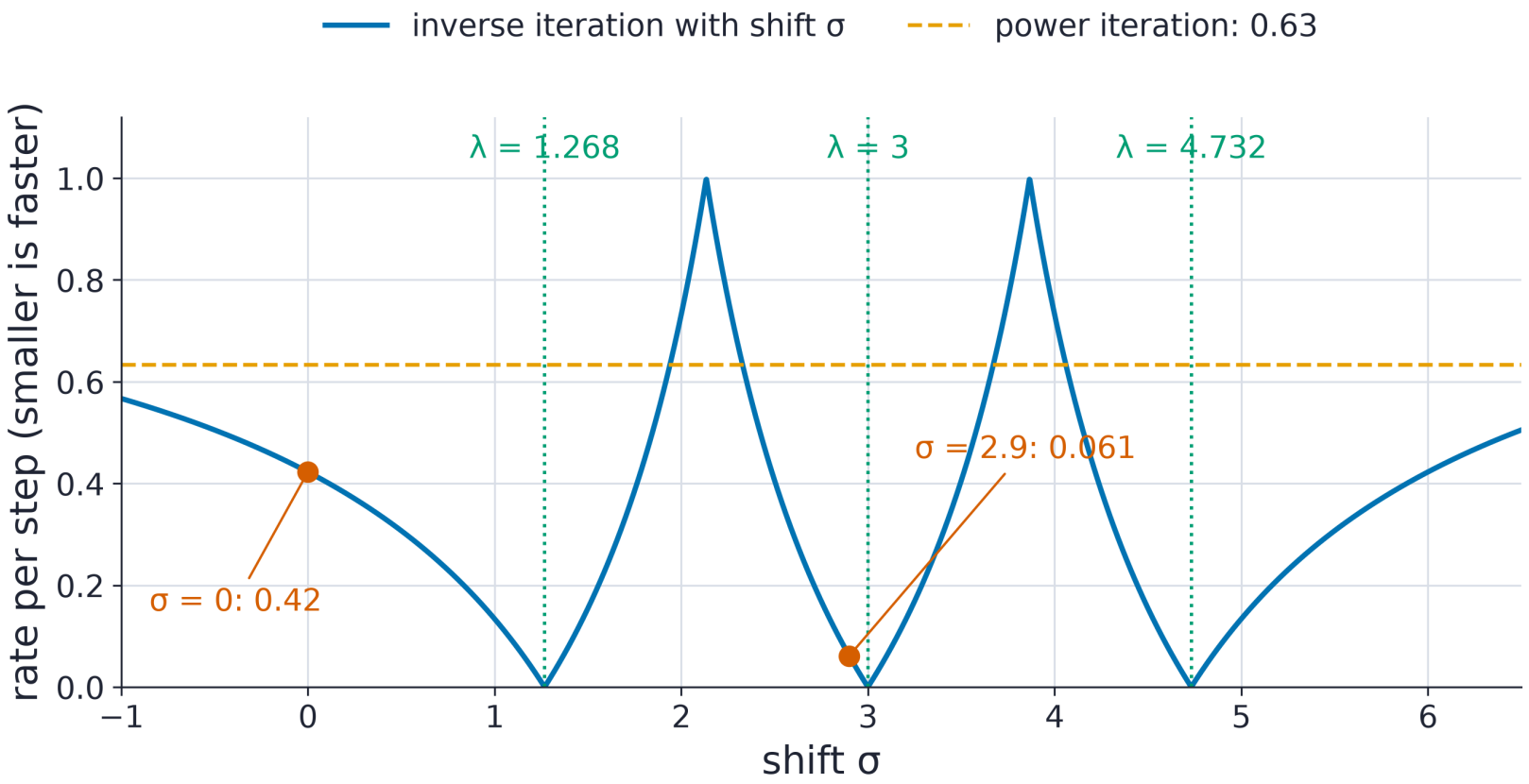
- $(A - \sigma I)^{-1}$ has eigenvalues $1/(\lambda - \sigma)$, same eigenvectors
- the λ nearest σ gives the largest $1/|\lambda - \sigma|$
- in code: LU-factor $A - \sigma I$ once, two triangular solves per step

In the demo

- Method: inverse iteration (shift σ)
- σ = 0 finds the smallest |λ|
- slide σ toward an eigenvalue: steeper slope
- σ halfway between two eigenvalues: rate 1
- σ an eigenvalue: A − σI is singular

Shift moves the eigenvalues, inversion turns the nearest one into the largest: power iteration on $(A - \sigma I)^{-1}$ reaches any eigenvalue.

Choosing the shift: the rate as a function of σ



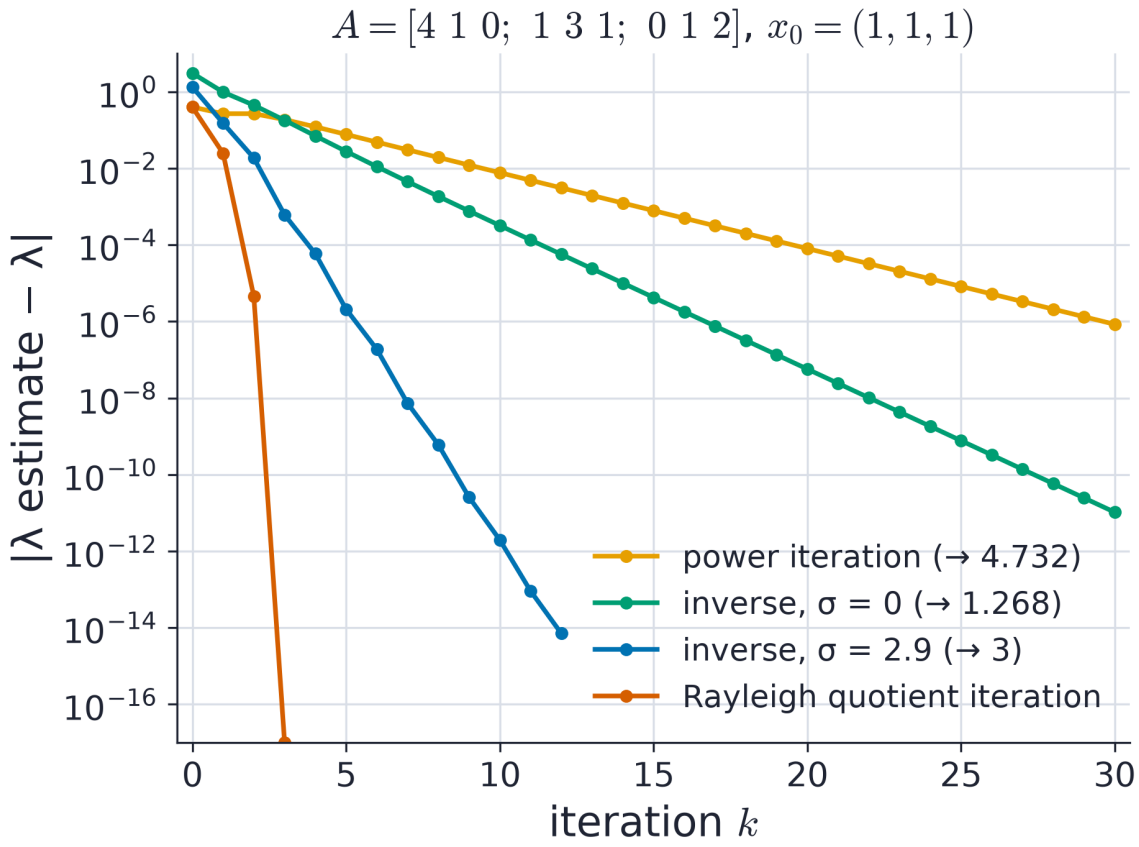
The rate drops to 0 at each eigenvalue and rises to 1 halfway between two of them. A shift close to the wanted λ is fast; a good guess is worth a lot.

Rayleigh quotient iteration: update the shift every step

$$\rho(x) = \frac{x^T A\, x}{x^T x}$$

$$\sigma_k = \rho(x_{k-1}), \qquad (A - \sigma_k I)\, y_k = x_{k-1}$$

k	RQI error
0	4.0×10^{-1}
1	2.5×10^{-2}
2	4.5×10^{-6}
3	$\approx 10^{-17}$ (roundoff)



Digits roughly triple each step (cubic convergence): the shift improves as the vector does, and the vector improves as the shift does.

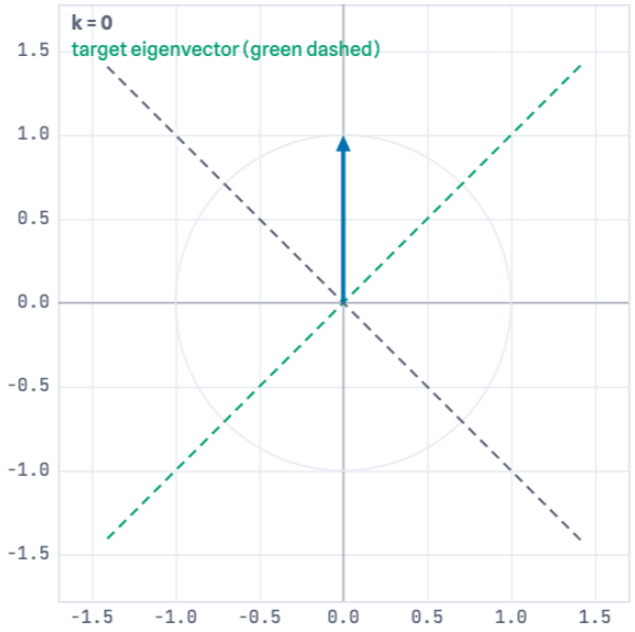
What the demo shows: power and inverse iteration

Eigenvalue Problems

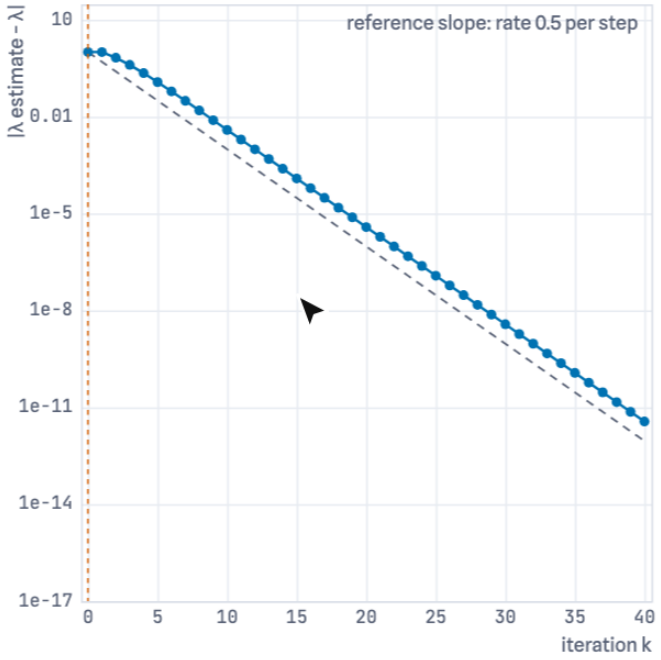
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Power and inverse iteration

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```
eigenvalues of A: 4, 2
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```

Common mistakes

“Power iteration is broken: it barely moves”

- Check $|\lambda_2/\lambda_1|$.
- At 0.95 it needs about 45 steps per digit.
- Shift, invert or use RQI.

Expecting power iteration to find every eigenvalue

- It finds the largest $|\lambda|$.
- Use a shift and inverse iteration for the others.

The estimate jumps once after converging ($\sigma = 2.9$, 3×3)

- The eigenvector $(-1, 1, 1)$ has tied largest entries of opposite sign, so the scale factor can flip sign.
- Stop at a tolerance or use $\rho(x)$.

Forming $(A - \sigma I)^{-1}$ in code

Factor $A - \sigma I = LU$ once, then solve each step.